Multiple ages of Nipissing Diabase intrusion: paleomagnetic evidence from the Englehart area, Ontario[1]

K. L. Buchan, K. D. Card, and F. W. Chandler
Geological Survey of Canada, 601 Booth St., Ottawa, Ont., Canada K1A 0E8

Received March 16, 1988
Revision accepted July 13, 1988

Nipissing Diabase sills and baked host sediments of the Coleman Member of the Huronian Supergroup east of Englehart, Ontario, are shown to have a characteristic remanent magnetization direction (called N3) that is steeply up and to the west ($D = 268.0°$, $I = -59.0°$, $k = 42$, $\alpha_{95} = 6.0°$). Petrographic study indicates that fresh pyroxene gabbro carries the N3 component at most sill sites. A baked contact test with the Coleman Member suggests that the magnetization is primary. The N3 magnetization direction is unlike either the N1 or N2 magnetization direction reported for Nipissing sills at other localities in the Southern Province. Three distinct ages of Nipissing sill emplacement are proposed. A single Nipissing sill site in the sampling area carries the N1 direction.

A northeast-trending diabase dyke crosscuts both the Nipissing sills and Coleman sediments. It carries an N2 direction and has overprinted nearby intrusive and sedimentary rocks ($D = 282.0°$, $I = 61.1°$, $k = 48$, $\alpha_{95} = 5.8°$). Several N3 sill sites far from the dyke may also carry a softer N2 overprint. However, the N3 and N2 directions and the direction of the present Earth's magnetic field fall near a great circle, making it difficult to separate the N2 and present-field components in such cases.

À l'est d'Englehart en Ontario, les filons-couches de la diabase de Nipissing et les roches sédimentaires encaissantes cuites du Membre de Coleman, du Supergroupe de l'Huronien, ont fourni une direction caractéristique d'aimantation rémanente (désignée N3) très rapprochée de l'ouest ($D = 268{,}0°$, $I = -59{,}0°$, $k = 42$, $\alpha_{95} = 6{,}0°$). L'examen pétrographique indique que, pour la majorité des sites, c'est le gabbro à pyroxène frais qui porte la composante N3. Le test du contact appliqué au Membre de Coleman suggère une aimantation primaire. La direction d'aimantation N3 diffère des deux directions d'aimantation N1 ou N2 qui ont été rapportées pour les filons-couches de Nipissing d'autres localités dans la province du Sud. Trois âges distincts de mise en place des filons-couches de Nipissing sont proposés. Dans la région échantillonnée, un seul site d'un filon-couche de Nipissing a fourni une direction N1.

Un dyke de diabase, de direction nord-est, recoupe les filons-couches de Nipissing ainsi que les sédiments de Coleman. Il porte une direction N2 et surimpressionne les roches intrusives et sédimentaires des alentours ($D = 282{,}0°$, $I = 61{,}1°$, $k = 48$, $\alpha_{95} = 5{,}8°$). Plusieurs sites éloignés de ce dyke, avec direction N3, peuvent également porter une surimpression N2, mais plus atténuée. Les directions N3 et N2 et la direction du champ magnétique terrestre actuel pointent près d'un grand cercle, dans ces cas la séparation des composantes de N2 et du champ actuel est rendue difficile.

[Traduit par la revue]

Can. J. Earth Sci. **26**, 427–445 (1989)

Introduction

Nipissing Diabase sills, with isotopic ages of about 2200 Ma, intrude sediments of the Early Proterozoic Huronian Supergroup in the Southern Province of the Canadian Shield (Fig. 1). Two distinctly different paleomagnetic directions, called N1 and N2, have previously been reported for Nipissing sills (Roy and Lapointe 1976). The N1 remanence is usually up and to the north (rarely down and to the south), whereas the N2 remanence is always steeply down and to the west.

Interpretation of the N1 and N2 components is controversial (see discussions in Irving and McGlynn 1981; McGlynn and Irving 1981). Symons and Londry (1975) suggested that N1 is primary and N2 is a viscous overprint. Roy and Lapointe (1976) interpreted N2 as the primary component and considered N1 to be secondary. Positive baked contact tests with the Huronian sediments were reported for both the N1 (Symons 1967; Morris 1979) and N2 (Roy and Lapointe 1976) components, leading Morris (1979) and Stupavsky and Symons (1982) to propose that two ages of Nipissing intrusions are present. McGlynn and Irving (1981) questioned the reliability of the Nipissing baked contact tests. Buchan and Card (1985) compared the petrographic and paleomagnetic characteristics of Nipissing sills from several localities and found no correlation between paleomagnetic direction and degree of alteration; both N1 and N2 remanences are carried by petrographically fresh gabbro. These observations support the hypothesis of two ages of intrusion.

Irving and McGlynn (1981) have suggested that the presence of much younger intrusions, petrologically difficult to distinguish from the Nipissing Diabase, may account for unusual Nipissing components, such as the one reported by Roy and Lapointe (1976) in the New Liskeard area. Roy and Lapointe (1976) observed this component, which has a shallow westward and up direction, near a fault and attributed it to local secondary overprinting.

In the Cobalt Plate (Fig. 1) Huronian sediments are gently folded and Nipissing intrusions are relatively fresh and undeformed (Card and Pattison 1973). Nipissing sills in the area east of Englehart, Ontario, were sampled in the present study to establish their direction of remanent magnetization. Huronian sediments of the Gowganda Formation (Coleman Member) were also studied, as was a 150 m wide northeast-trending diabase dyke that cuts both the Coleman sediments and Nipissing Diabase. It was hoped that these units would help to establish whether the sill remanence is primary or secondary.

Geological setting

Regional geology

The Huronian Supergroup of the Southern Province is a

[1]Geological Survey of Canada Contribution 51287.

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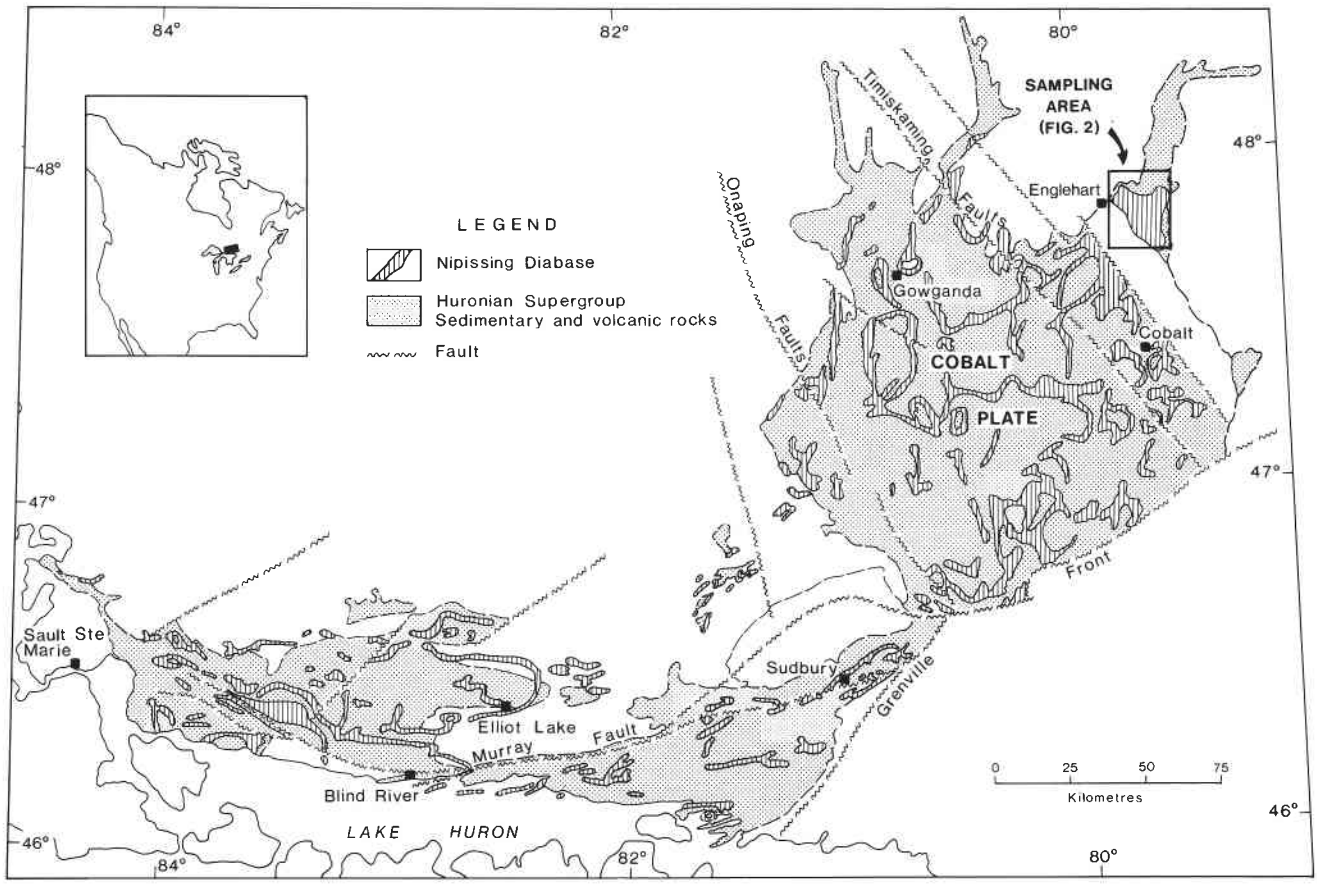


FIG. 1. Geology of the Southern Province. The Murray Fault is approximately the boundary between the relatively highly metamorphosed Sudbury – Blind River zone and the little metamorphosed Sault Ste. Marie – Cobalt zone. The study area is enclosed by a rectangle.

wedge of Early Proterozoic clastic sediments with local basal volcanic accumulations. It lies unconformably on the southern flank of the Archean Superior Province craton and increases in thickness southward to a maximum of about 12 km. The age of the supergroup is bracketed by U–Pb zircon and baddeleyite ages of 2490–2450 Ma for lower Huronian volcanics and gabbro-anorthosite intrusions (Krogh *et al.* 1984) and 2219 ± 4 Ma for Nipissing Diabase intrusions that cut the supergroup (Corfu and Andrews 1986). The Huronian consists of three older units, the Elliott Lake, Hough Lake, and Quirke Lake groups, which occur south of latitude 47°N, and a younger unit, the Cobalt Group, which overlaps the older groups and Archean basement and extends northward to about latitude 48°N, where it forms the roughly circular Cobalt Plate (Frarey and Roscoe 1970; Card *et al.* 1972; Fig. 1).

In areas south of the Murray fault (Fig. 1), Huronian rocks and Nipissing intrusions are folded and variously metamorphosed under greenschist to lower amphibolite conditions of the low-pressure intermediate facies series. North of the Murray fault, in the Sault Ste. Marie – Elliot Lake area, and in the Cobalt Plate, they are little deformed and unmetamorphosed, or only deuterically altered. There is evidence for deformation of the Huronian prior to emplacement of Nipissing Diabase intrusions, but the major deformation and metamorphism are probably correlative with the Early Proterozoic Penokean Orogeny that affected the entire Southern Province about 1850 ± 50 Ma ago (Van Schmus 1976; Card 1978). Tectonic settings proposed for deposition of the Huronian Supergroup include graben-style basins (Card 1978), an aulacogen (Young 1983), and a consuming continental margin (Van Schmus 1976). Zolnai *et al.* (1984) proposed that the Huronian is a passive margin sequence deposited during rifting and ocean basin formation along the southern margin of the Superior craton. Collisional tectonism about 1850 Ma ago resulted in northward thrusting, deformation, and metamorphism of the Huronian and Nipissing rocks.

The Cobalt Group comprises four mutually conformable formations, in ascending order, the Gowganda (conglomerate, argillite), Lorrain (sandstone), Gordon Lake (sandstone, argillite), and Bar River (sandstone) formations. The glacigene Gowganda Formation (Lindsey 1969; Young 1970), which with the Lorrain Formation dominates exposure of the Cobalt Group in the Cobalt Plate, is up to 2700 m thick. The Gowganda Formation is subdivided into a lower Coleman Member, a conglomerate-rich part attributed to deposition in a shallow marine environment, and an upper Firstbrook Member, a fine-grained, upward-coarsening part attributed to a prograding deltaic regime (Thomson 1957; Rainbird 1985; Mustard and Donaldson 1987).

The Huronian Supergroup is intruded by sills, dykes, and cone sheets of tholeiitic gabbro of the Nipissing Diabase. The sills range in thickness from a few metres to several hundred metres. Isotopic ages of 2134 ± 40 Ma[2] (Rb–Sr biotite)

[2]Rb–Sr ages recalculated with $\lambda = 1.42 \times 10^{-11}$ a^{-1} (Steiger and Jäger 1977).

(Van Schmus 1965), 2116 ± 27 Ma[2] (Rb–Sr whole rock) (Fairbairn *et al.* 1969), and 2219 ± 4 Ma (U–Pb baddeleyite) (Corfu and Andrews 1986; Andrews *et al.* 1986) have been obtained for Nipissing sills from the Cobalt Plate, while an age of 2109 ± 80 Ma[2] (Rb–Sr whole rock) (Van Schmus 1965) was reported for sills of the Blind River – Bruce Mines area. Descriptions of the petrography and geochemistry of the Nipissing intrusions are given for the Sudbury area by Card (1978) and others, for the Maple Mountain area by Card *et al.* (1970), and for the Cobalt area by Hriskevitch (1968), Jambor (1971), and others, and are summarized by Card and Pattison (1973).

Nipissing diabase intrusions consist mainly of pyroxene gabbro and hornblende gabbro, but differentiation has led to the formation of minor amounts of feldspathic pyroxenite, gabbroic anorthosite, and granophyre. Olivine is present in several intrusions in the Cobalt and Englehart areas but the typical mafic minerals are augite, hypersthene, hornblende, and minor biotite. Calcic plagioclase, quartz, sulphides, iron–titanium oxides (ilmenite – titaniferous magnetite and ilmenite–hematite), and minor apatite form the remainder. Major- and minor-element chemistry indicates that all of these rocks formed from magma of similar composition with differentiation mainly by gravitative settling of early-formed mafic minerals with consequent depletion of residual liquids in Mg and Ca and enrichment in iron and alkalis.

Alteration of the Nipissing Diabase, whether by low-grade regional metamorphism as in the Sudbury area (Card 1978) or by deuteric alteration as in the Cobalt area, resulted in replacement of the primary minerals by uralitic amphiboles, albite, epidote, chlorite, and quartz or quartz–feldspar intergrowths. The primary ilmenite–magnetite and ilmenite–hematite intergrowths are replaced by magnetite–sphene–leucoxene aggregates. In addition, uralitization of the mafic minerals commonly results in formation of fine-grained oxides and sulphides.

Also cutting the Huronian rocks are northeast-trending quartz diabase dykes of the Preissac swarm with an $^{40}Ar-^{39}Ar$ isotopic age of 2150 Ma (Hanes and York 1979) and younger northeast trending olivine diabase dykes of the Abitibi swarm with a U–Pb baddeleyite age of 1140 Ma (Krogh *et al.* 1987). The older Preissac dykes are similar to the Nipissing intrusions in chemistry, petrography, and isotopic age.

Geology of the study area

The study area (Fig. 2) lies just east of Englehart in the northern portion of the Cobalt Plate where rocks of the Huronian Supergroup are faulted but otherwise little deformed and unmetamorphosed. The geology of the area is described in detail by Lovell and Ploeger (1980). They note that "variations in dips in the Cobalt Group rocks are controlled predominantly by the Archean bedrock topography and differential rates of compaction of the sedimentary rocks rather than by tectonism."

Basal conglomerate and overlying red and grey argillite to sandstone of the Coleman Member outcrop along the shores of Wendigo Lake. The sediments are cut by Nipissing diabase sills, which extend over a broad area to the south of Wendigo Lake. The sills and sediments are cut by a 150 m wide northeast-trending diabase dyke. These relationships provide opportunities for testing the primary nature of the magnetic remanence in the Coleman sediments (conglomerate test), the Nipissing sills (baked contact test with surrounding sediments), and the northeast-trending dyke (baked contact test with sills or sediments).

Phanerozoic normal movements on faults of the Timiskaming fault system (Fig. 1) produced the Timiskaming Graben, the northward extension of the Mesozoic Bonnechere graben system (Lovell and Caine 1970).

Sampling and laboratory procedure

An average of five oriented drill cores or block samples were collected at each of 37 sites in the area east of Englehart (Fig. 2). Sixteen sites are in Nipissing Diabase, 15 in Coleman sediments, and 6 in a northeast-trending diabase dyke that cuts Coleman sediments and Nipissing Diabase. To carry out baked contact tests, some sampling sites were located along horizontal profiles roughly at right angles to intrusive contacts. Distances to the contact were measured along the present horizontal. For a vertical dyke this is the distance perpendicular to the contact, but for sills this merely represents a maximum possible distance perpendicular to the contact.

Two or three cylindrical specimens (2.5 cm diameter, 2.2 cm length) were cut from each core or block sample. Specimen magnetizations were measured on a Schonstedt spinner magnetometer.

At each sill and dyke site, two specimens per sample were alternating-field (AF) demagnetized in stepwise fashion to peak fields of 2.5, 5.0, 10, 15, 20, 25, 30, 35, 40, 50, 60, 70, 80, 90, and 100 mT using a Schonstedt AF demagnetizer. Selected specimens from these sites were thermally demagnetized at peak temperatures of 200, 300, 400, 450, 500, 530, 550, 560, 570, and 580°C using a Schonstedt thermal demagnetizer. Occasionally, more detailed step heating was carried out.

At some sill sites stable end points were obscured in the thermal demagnetization experiments by a softer magnetization component, which may be a viscous remanent magnetization (VRM) or the softer of two stable components. Therefore, to uncover the blocking temperature spectrum of the high-temperature component, a two-stage cleaning procedure (Park 1975) was adopted. This involved stepwise AF demagnetization to eliminate the softer components before the stepwise thermal cleaning was begun.

Stepwise AF demagnetization to 100 mT was found to be useful for some sedimentary sites, especially those located close to dyke or sill contacts where magnetite usually carries the stable remanence. At other sites the remanent magnetization was hardly affected by this procedure and, hence, thermal or two-stage AF–thermal demagnetization was employed. Since hematite usually carries the stable magnetization at these other sites, the following peak temperatures were used: 200, 300, 400, 450, 500, (530), 550, 580, 600, 620, 640, 660, and 680°C. Additional steps were included when appropriate.

A polished thin section was examined for each paleomagnetic sampling site. Of particular interest was the correlation between opaque mineralogy and paleomagnetic characteristics.

Paleomagnetic results

Paleomagnetic directions observed at each of the sampling sites of the Englehart area are summarized in Figs. 2 and 3. Structural corrections have not been made unless otherwise noted. Results are described in order of decreasing age of the geological units involved.

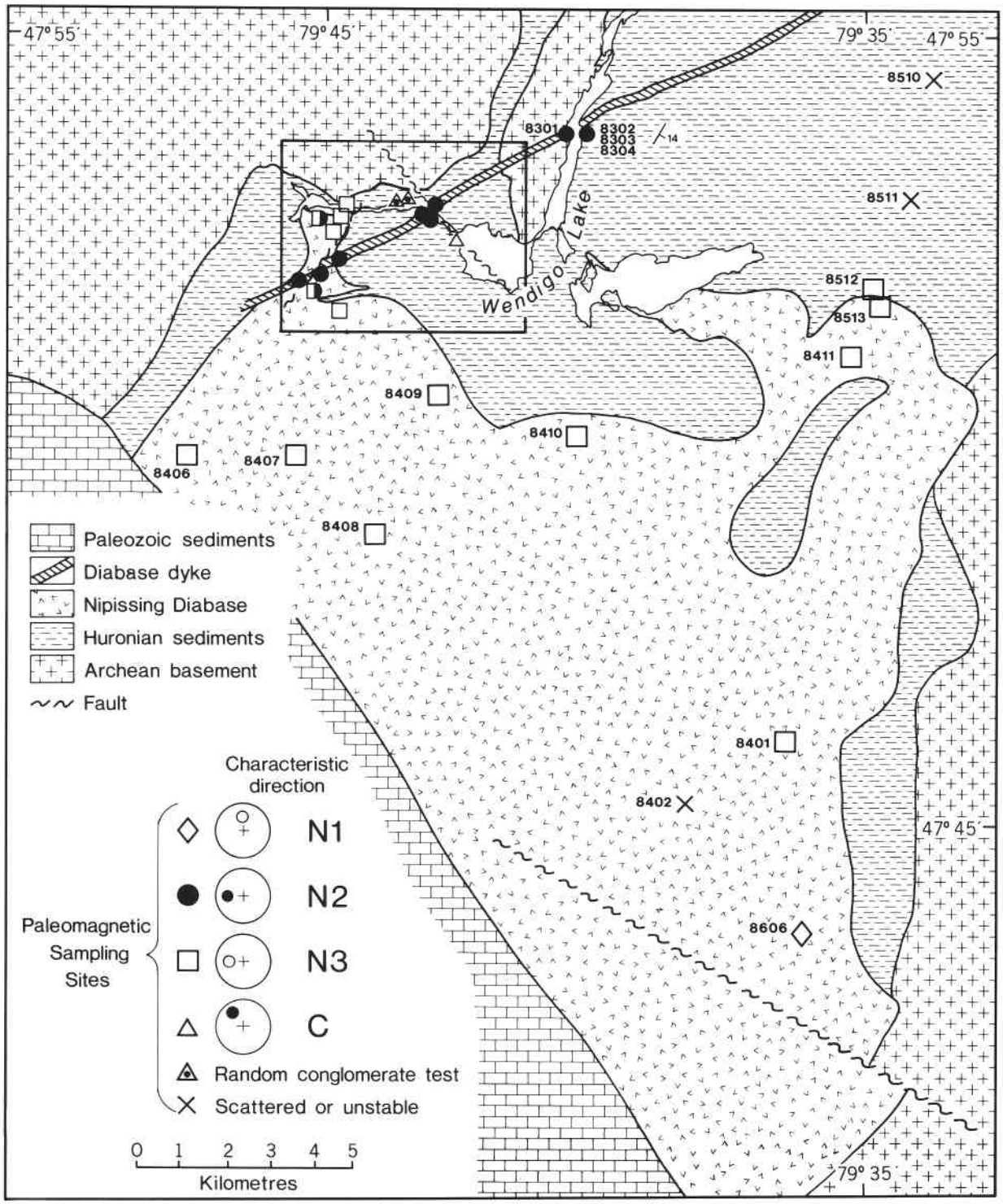


FIG. 2. Geology, sampling sites, and paleomagnetic summary for the Englehart area. Geology after Ontario Geological Survey Preliminary Map P. 1198 (revised). As described in the text, some sites with the characteristic N3 remanence may also carry a softer N2 component that is difficult to separate from a viscous magnetization directed along the present Earth's magnetic field. The congested area near the western arm of Wendigo Lake (rectangle) is enlarged in Fig. 3.

Coleman sediments

Sixteen sites were sampled in gently folded sediments of the Coleman Member of the Gowganda Formation. Eleven of these sites are located close enough to Nipissing sills or to the northeast-trending diabase dyke to have been magnetically overprinted at the time of sill or dyke intrusion. Results from the overprinted sites will be discussed below with the sill and dyke data. Five sedimentary sites are not overprinted. One is located in redbeds (8310), two in grey sediments (8510 and 8511), and two in the basal conglomerate (8313 and 8314).

Site 8310 is the only redbed site sufficiently far from the sills and dyke (Fig. 3) that it was not magnetically overprinted at the time of intrusion. AF cleaning removes a component that is directed steeply down and is probably of viscous origin.

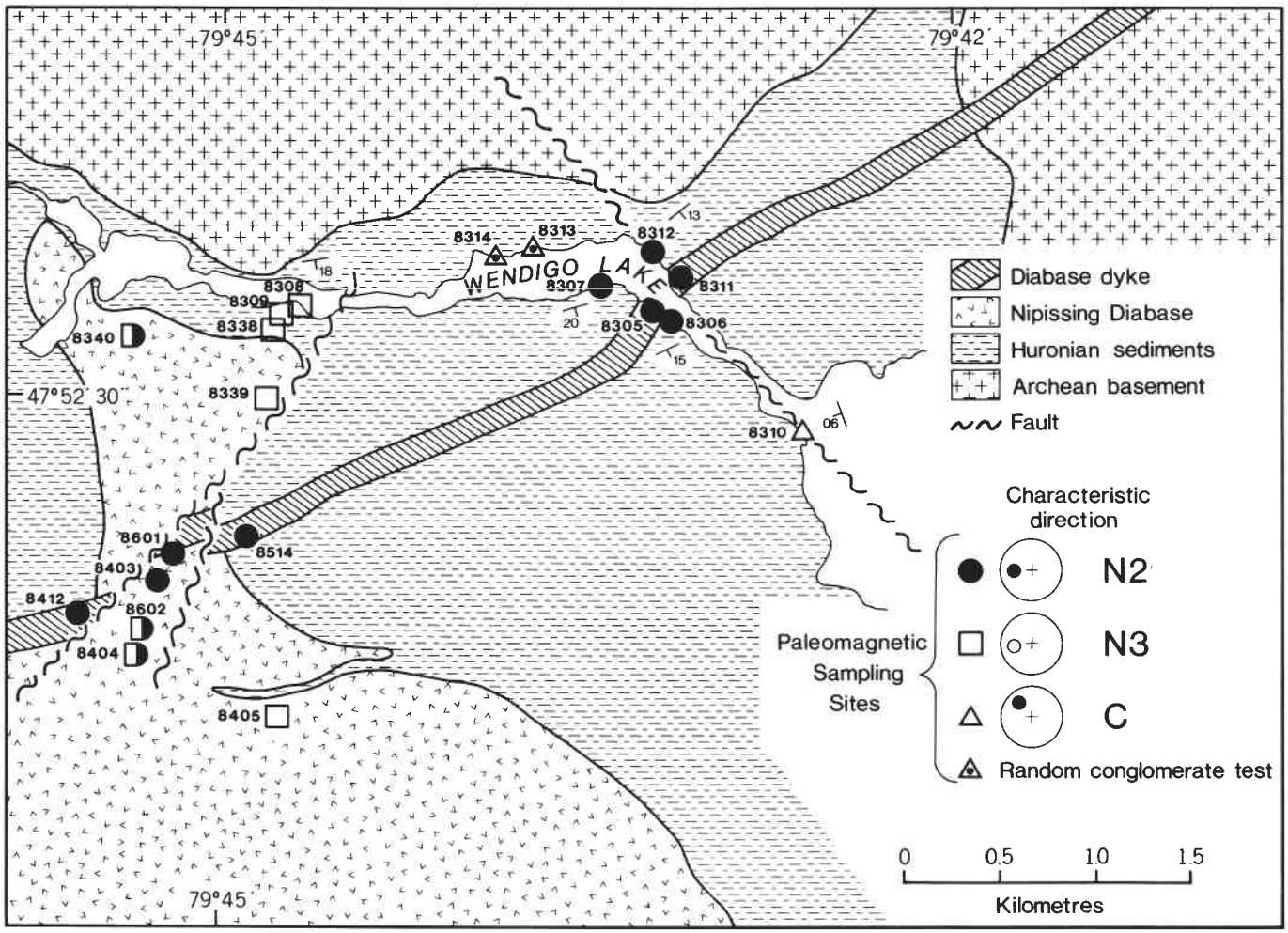


FIG. 3. Geology, sampling sites, and paleomagnetic summary for the western arm of Wendigo Lake. Geology modified after Ontario Geological Survey Preliminary Map P. 1198 (revised).

With subsequent thermal demagnetization (Fig. 4) a shallow northwestward and down direction is attained. This magnetization will be called the C remanence. Coercivities of the C component are greater than 100 mT and most blocking temperatures are greater than 600°C, indicating that hematite is the remanence carrier. The stable magnetization direction observed at site 8310 is unlike those observed in either the sills or dyke.

Samples were collected from 14 mafic cobbles of two basal conglomerate sites (8313 and 8314). The cobbles are derived from Archean igneous rocks. Thermal demagnetization of three specimens from site 8313 demonstrates that the stable magnetization components are carried by low-titanium titanomagnetite with blocking temperatures near 580°C. Both sites are located relatively far from the sills and dyke and show no evidence of overprinting. On AF cleaning, remanence directions of the cobbles at these sites (Fig. 5) are stable but highly scattered. The 14 remanence directions are shown to be statistically random (Watson 1956), since the length of the resultant vector, $R = 3.6$, is less than $R_0 = 6.0$, where R_0 is the value of R that will be exceeded with a probability of 5% in sampling a random population.

This positive conglomerate test suggests that the magnetization of Coleman redbed site 8310, described above, has not been thermally overprinted. Nevertheless, since only one site is available, and because no further redbed sites with proper structural control were located far from the sills and dyke, no attempt will be made to estimate a paleopole or paleolatitude for the Coleman redbeds.

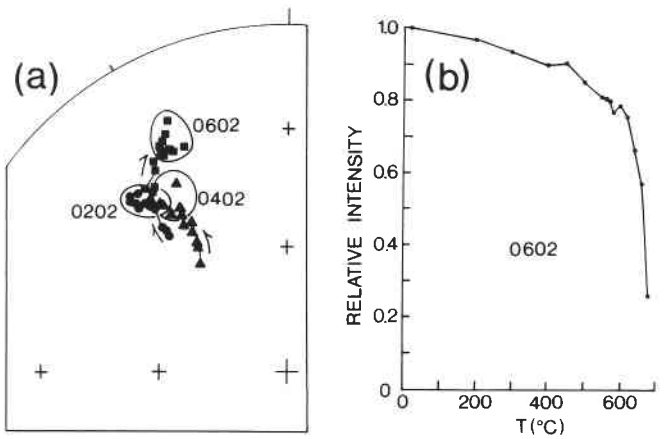


FIG. 4. Thermal demagnetization results for Coleman redbed site 8310 following AF demagnetization to 100 mT: (*a*) directional data are plotted with respect to the paleohorizontal on an equal-area net with closed symbols indicating down directions; (*b*) intensities on thermal demagnetization are normalized to the intensity at 100 mT.

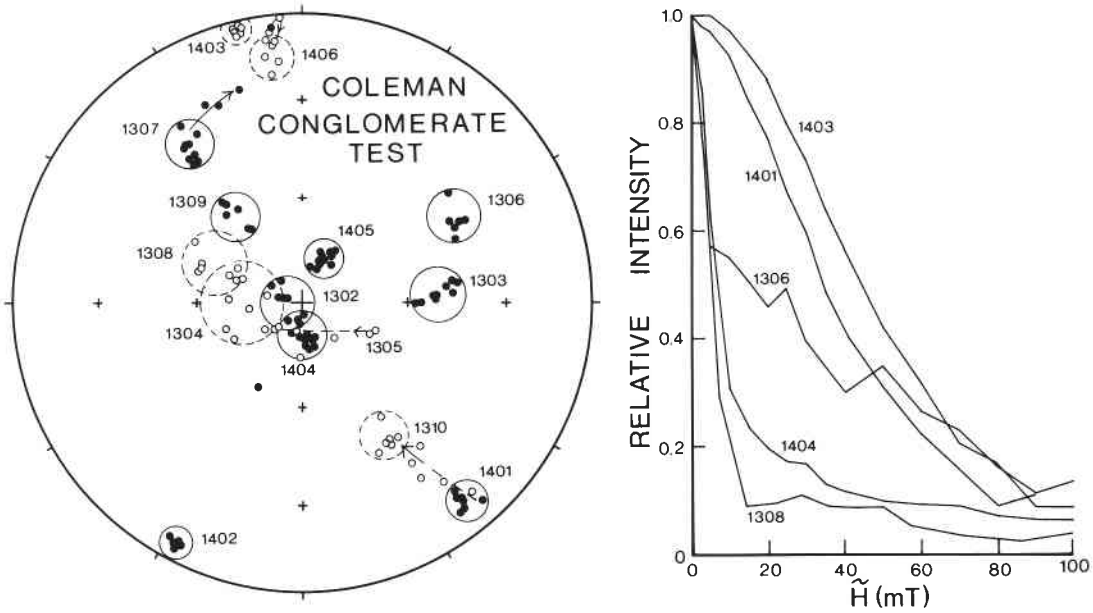


FIG. 5. AF demagnetization directional and intensity results for mafic cobbles of Coleman conglomerate sites 8313 and 8314. Closed (open) symbols indicate down (up) directions. Directional data are plotted with respect to the present horizontal.

Grey sediments of the Coleman Member (sites 8510 and 8511 of Fig. 2) are not magnetically stable except where altered chemically by a nearby intrusion (see discussion of site 8512 below). Hence the grey units are of no use in establishing a primary Coleman paleopole.

Nipissing Diabase and contact rocks

Nipissing Diabase from outside the Englehart area commonly carries an N1 magnetization (directed either north and up at about 40° or south and down at 40°) or an N2 magnetization (directed steeply down to the west). In the area of the present study, however, the dominant magnetization of Nipissing sills is steeply up and to the west. This direction will be referred to as N3.

As summarized in Fig. 2, the N3 magnetization occurs at all Nipissing sites in the sampling area except unstable site 8402, site 8606, which is the most southerly site and carries the N1 direction, and sites close to the northeast-trending diabase dyke that crosscuts the sills. Upon stepwise AF demagnetization a stable magnetization direction is usually attained in a field of 35 mT or less. Examples from several N3 sites are illustrated in Figs. 6*a*–6*g*. The steeply up west end point is maintained to the highest AF's that were employed (100 mT). Even after demagnetization in peak fields of 100 mT, many specimens still retain a relatively high percentage of their original natural remanent magnetization (Fig. 6*i*). The single N1 site, 8606 (Fig. 6*h*), has a stable up and north direction.

Stepwise thermal demagnetization (Figs. 7*a*, 7*b*) demonstrates that blocking temperatures of the N3 component are usually close to the Curie temperature of magnetite (Fig. 7*c*). Softer components, usually directed steeply down, have a broad range of blocking temperatures from room temperature to >500°C. To eliminate the softer components and better examine the blocking-temperature spectrum of the N3 component, specimens from each site were demagnetized in alternating fields of 100 mT or less before the thermal demagnetization procedure was begun. Thermal portions of typical two-stage cleaning experiments are shown in Figs. 7*d*, 7e. AF demagnetization completely eliminates the steeply downward components and a stable N3 end point is maintained during the thermal portion of the run. N3 blocking temperatures between 500 or 550°C and 580°C indicate that low-titanium titanomagnetite carries the N3 component.

To analyse the softer components at N3 sites, orthogonal plots of horizontal and vertical magnetization components were prepared for each specimen (Fig. 8). A straight-line segment through a range of AF's or temperatures indicates the elimination of a single magnetization component.

An unusual example is specimen 84110401 (Fig. 8*a*), which has no component softer than the N3 remanence. More often a soft viscous component with a steeply down direction close to the present Earth's magnetic field (PEF) is observed (Fig. 8*b*). In a few cases the viscous component is unrelated to the direction of the present Earth's field (Fig. 8*c*) and may have been acquired during laboratory storage.

In many specimens at N3 Nipissing sites there is evidence of an N2 component (steeply down and to the west) with coercivities and blocking temperatures intermediate between those of the VRM and N3 components (Figs. 8*d*–8*g*). It is often difficult to distinguish between an N2 component and a combination of VRM and N3, since the N2 direction lies near the PEF–N3 great circle. Nevertheless, in some specimens the N2 component can be clearly seen on the orthogonal component plot. For example, in specimen 84070401 (Fig. 8*d*) a soft viscous component is eliminated between 0 and 2.5 mT, an N2 component between 2.5 and 20 mT, and an N3 component above 20 mT. Figure 8*g* illustrates a specimen with the superimposed N2 component on thermal demagnetization. In specimen 84050203 a VRM is erased between room temperature and 530°C, an N2 component between 530 and 570°C, and an N3 component between 570 and 590°C.

AF demagnetization consistently eliminates the N2 magnetization in a field of less than 35 mT. Although continued AF cleaning to 100 mT does not completely erase the N3 com-

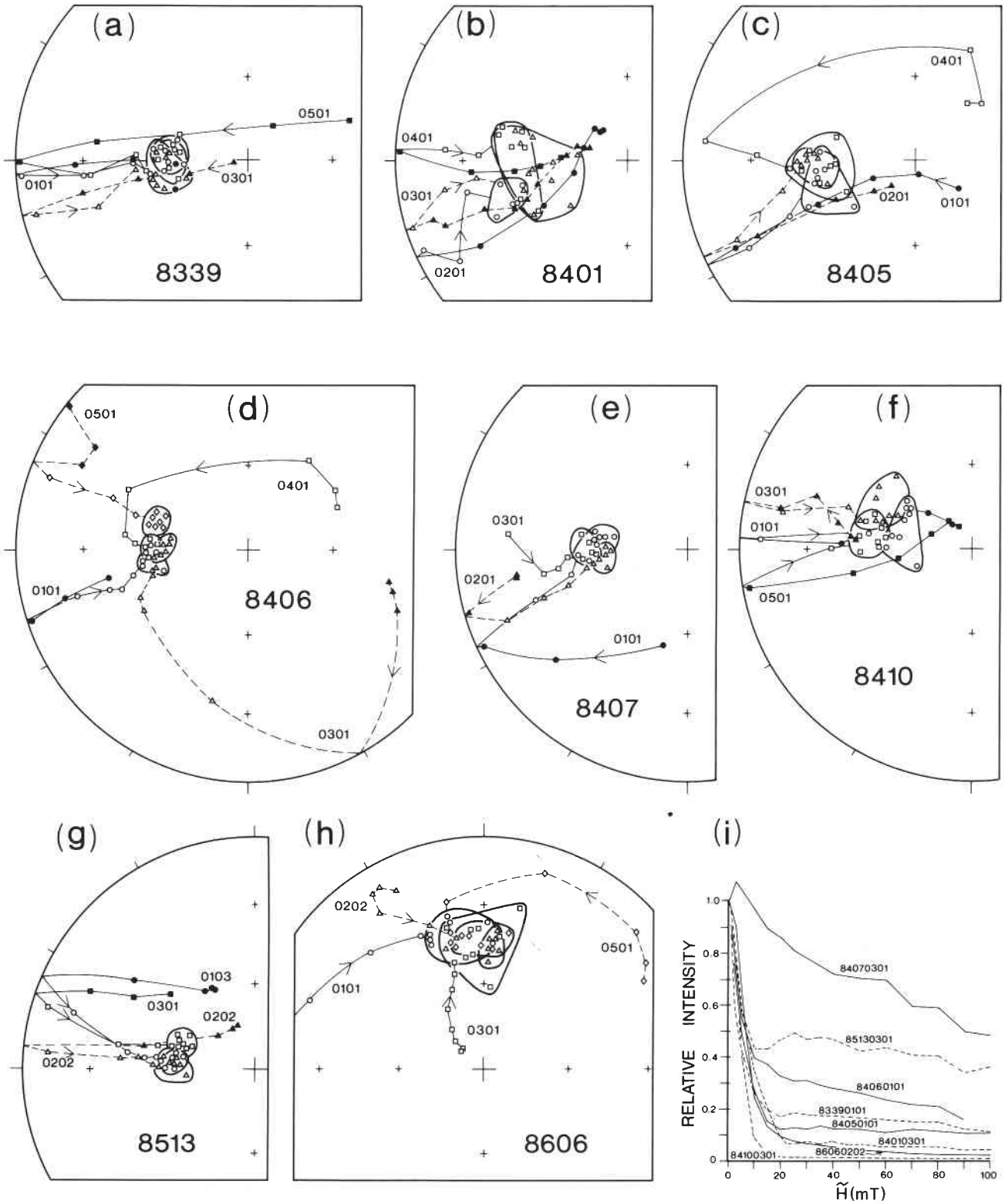


FIG. 6. AF demagnetization results with respect to the present horizontal for typical Nipissing sill sites far from the northeast-trending dyke: (*a*)−(*g*) sites carrying the N3 magnetization; (*h*) the single site carrying the N1 magnetization; (*i*) relative intensity decay.

ponent (Figs. 8*d*−8*f*), subsequent thermal demagnetization demonstrates the single-component nature of the remaining (N3) remanence (Figs. 8*h*, 8*i*).

To test the primary or secondary nature of the magnetization of the Nipissing sills two baked contact tests were conducted across the contact between Nipissing Diabase and Coleman sediments.

In the northeastern portion of the study area (Fig. 2), sampling sites were located in the sill (sites 8411 and 8513), in grey Coleman sediments within a few metres of the sill (site

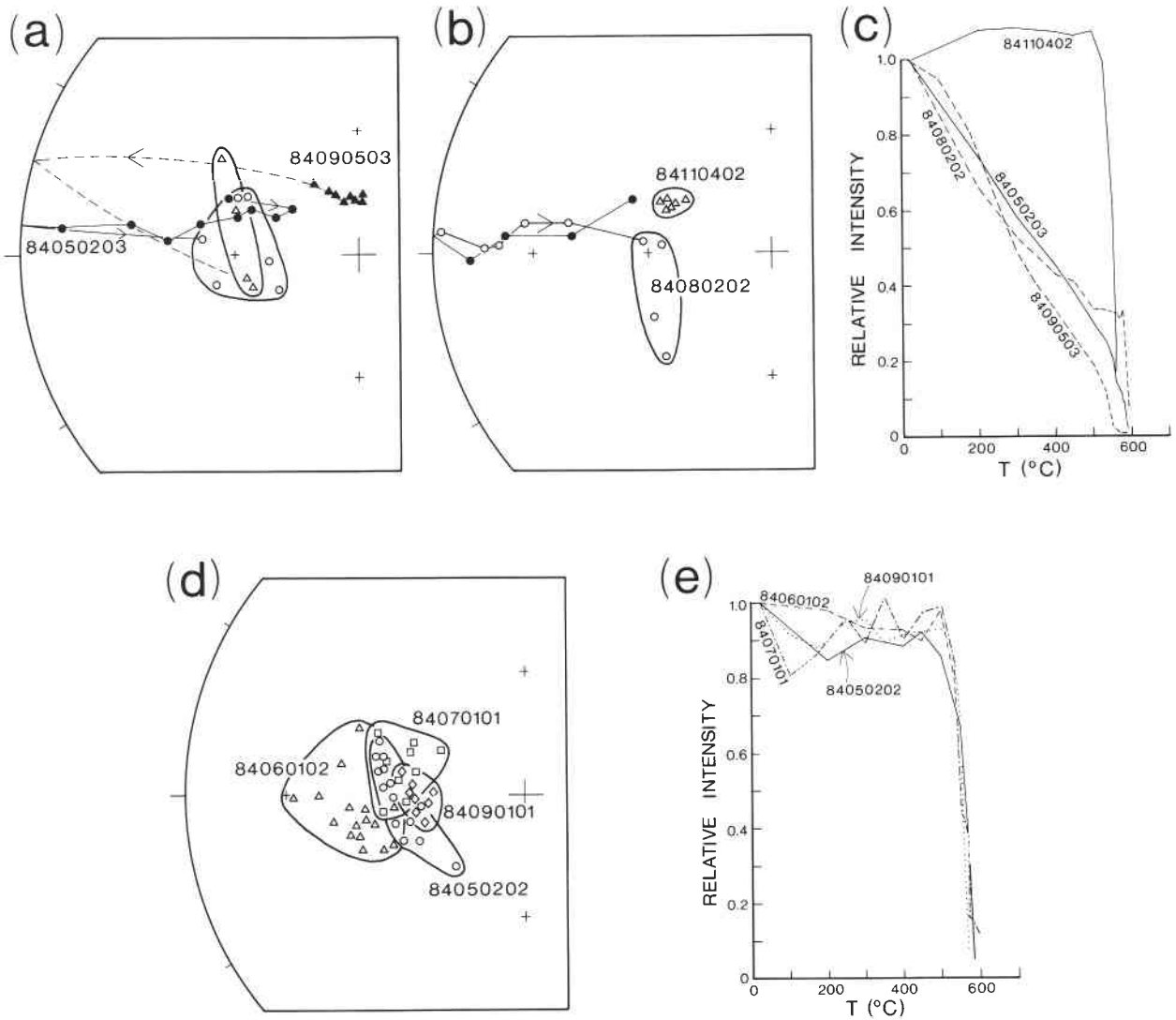


FIG. 7. Demagnetization results with respect to the present horizontal for typical N3 Nipissing sill samples: (*a*)–(*c*) thermal demagnetization; (*d*), (*e*) thermal demagnetization following AF cleaning to 100 mT. In (*e*) intensities are normalized to the intensity obtained after demagnetization in AF's of 100 mT.

8512), and in grey sediments at distances of 2 and 5 km from the sill (site 8511 and 8510 respectively). The sites in the Nipissing sill (Figs. 6*g*, 8*a*, 8*b*) and in adjacent sediments (see specimen 85120301 of Fig. 9*a*) carry a stable N3 magnetization, while sites 8511 and 8510 far from the sill are unstably magnetized. Perhaps the grey sediments of this area were not stably magnetized before Nipissing intrusion, but chemical alterations at the time of intrusion produced magnetic minerals capable of carrying a stable remanence in sediments adjacent to the sill (i.e., site 8512). Magnetic blocking temperatures at site 8512 indicate that magnetite carries the N3 overprint.

On its own this baked contact test is negative, because sites far from the sill are unstably magnetized. It does not conclusively demonstrate the primary nature of the N3 sill magnetization, since long after sill emplacement a regional metamorphic event could have thermally reset the stable remanence of the sill and adjacent sediments without chemically altering distant sites 8510 and 8511 to produce magnetically stable minerals. Nevertheless, such a thermal overprint appears unlikely, since petrographic evidence (below) indicates that the sill is relatively unmetamorphosed.

A second baked contact test was carried out near the western arm of Wendigo Lake (Fig. 3). Sampling sites 8339 and 8340 are located in the Nipissing sill. They carry the characteristic N3 magnetization (Fig. 6*a*), although some specimens, especially those at site 8340, also carry the softer N2 component.

Sedimentary sites 8338, 8309, and 8308 were collected at horizontal distances of about 15, 100, and 150 m, respectively, from the sill contact. All three sites have an up and west N3 magnetization with considerable scatter between sites. Examples of directional characteristics upon AF demagnetization are illustrated in Figs. 9*a*–9*c*. Thermal demagnetization (Figs. 9*d*–9*f*) demonstrates that the N3 remanence is carried by magnetite at site 8338 close to the sill and by hematite at sites 8308 and 8309 farther from the sill. Apparently, hematite was retained in the redbeds of sites 8308 and 8309, while hematite was replaced by magnetite at site 8338 at the time of sill intrusion. The N3 overprint at sites 8308 and 8309 is, therefore, undoubtedly of thermal origin (see further discussion in petrography section).

Unfortunately, the Coleman redbeds do not outcrop directly across the lake from site 8308. However, about 1 km farther

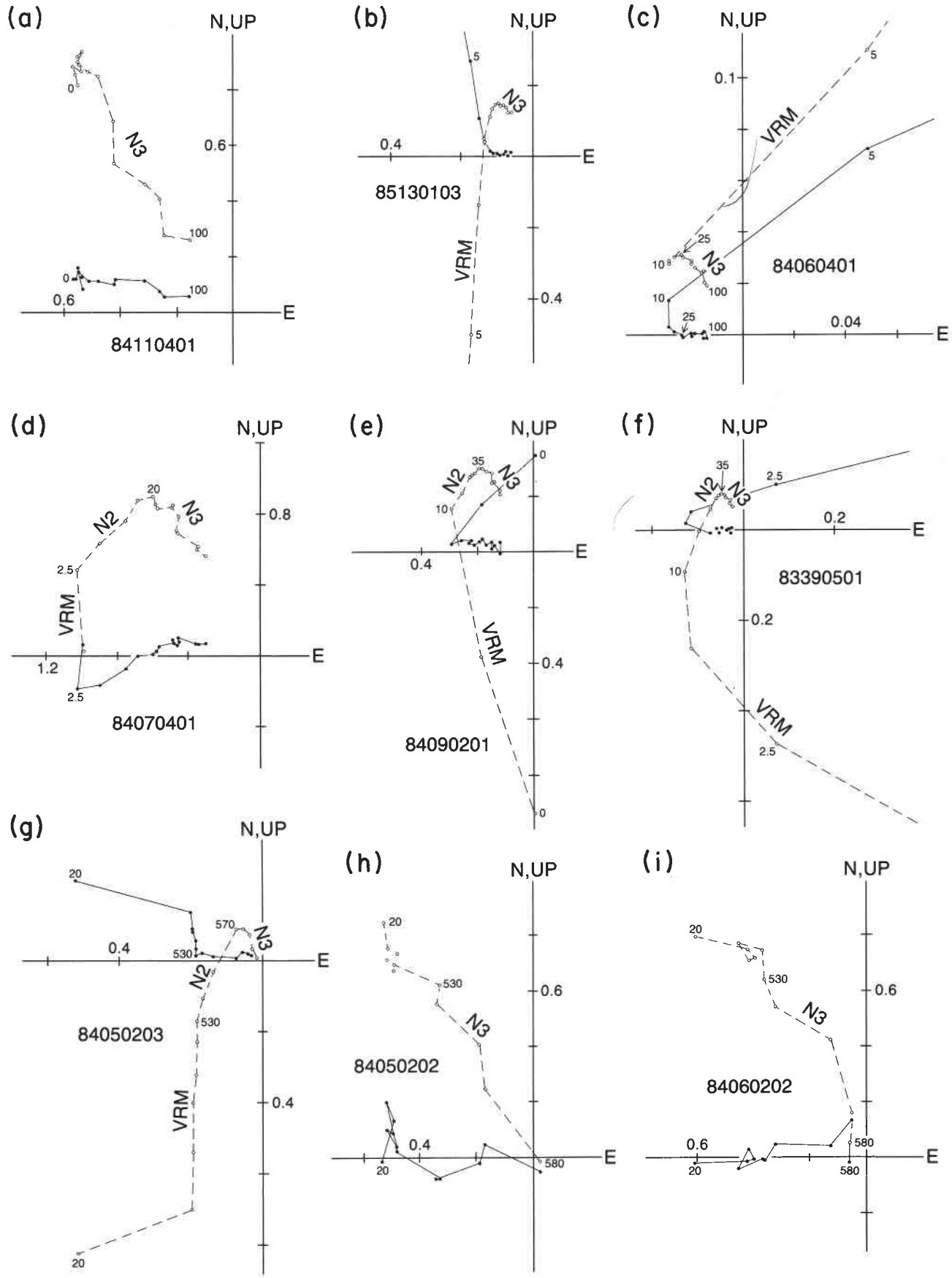


FIG. 8. Orthogonal-component plots with respect to the present horizontal for typical N3 Nipissing sill specimens: (*a*)–(*f*) AF demagnetization with AF's in mT; (*g*) thermal demagnetization; (*h*), (*i*) thermal demagnetization following AF cleaning to 100 mT. Closed symbols indicate projection onto the horizontal plane; open symbols indicate projection onto the up and east vertical plane.

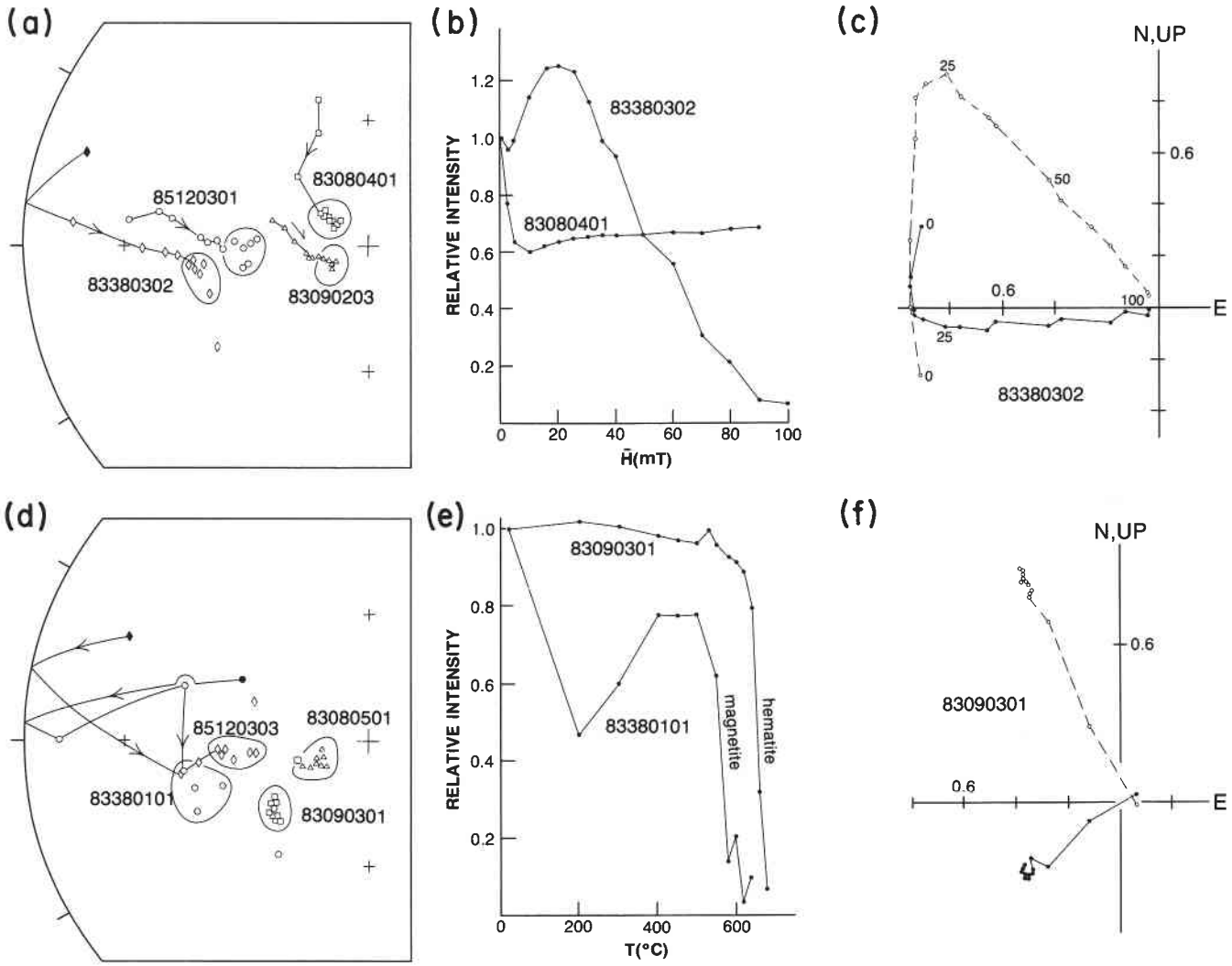


FIG. 9. Typical results with respect to the present horizontal for baked sediments close to Nipissing sills: (*a*)–(*c*) AF demagnetization; (*d*)–(*f*) thermal demagnetization (specimens 83380101 and 85120303) or thermal demagnetization following AF demagnetization to 500 mT.

east (see Fig. 3) mafic cobbles of a basal conglomerate of the Coleman Member (sites 8313 and 8314) carry stable but random directions of magnetization as described earlier. This positive conglomerate test shows that the remanence of the conglomerate cobbles was not significantly affected by intrusion of the Nipissing sill. Even farther east, Coleman redbed site 8310 (see Fig. 3) carries a stable pre-sill magnetization.

These results constitute a positive baked contact test, indicating that the N3 magnetization was acquired at the time of Nipissing intrusion. Nearby sedimentary sites (8338, 8309, and 8308) were remagnetized in the N3 direction, but conglomerate sites (8313 and 8314) and redbed site 8310 farther from the sill were not overprinted.

If the N3 component of the Nipissing Diabase is primary, then the lower coercivity, lower blocking temperature N2 component occurring at some N3 sites must have been acquired at a later time. The origin of the N2 overprint will be considered later.

A single Nipissing sill site (8606), the southernmost site in the sampling area (Fig. 2), carries a stable N1 magnetization. This component, directed up and north, is illustrated in Fig. 6*i*. Like the N3 magnetization of other sill sites, the N1 remanence at site 8606 is carried by magnetite. There is no evidence at this site of superimposed N2 or N3 components.

Northeast dyke and contact rocks

Samples were collected at six sites over 7 km of the 150 m wide diabase dyke that cuts Coleman sediments and Nipissing sills near Wendigo Lake (Figs. 2 and 3). At each site a stable N2 magnetization direction is observed on AF or thermal cleaning (Figs. 10*a*, 10*b*). Blocking temperatures are near 580°C, indicating that magnetite carries the stable magnetization.

The N2 magnetization is similar in direction to that reported for dykes of the northeast-trending 1140 Ma Abitibi swarm (Ernst *et al.* 1987). On the other hand, N1 and N2 directions have both been reported for northeast-trending dykes of the older 2150 Ma Preissac swarm (Larochelle 1966). Therefore the occurrence of the N2 direction in the Englehart dyke does not establish whether the dyke belongs to the Preissac or the Abitibi swarm.

A baked contact test was conducted between the dyke and host sediments where the dyke crosses Wendigo Lake (Fig. 3). Farther to the southwest, where the dyke cuts Nipissing sills, a second contact test was attempted.

At the locality of the first baked contact test, two dyke sites

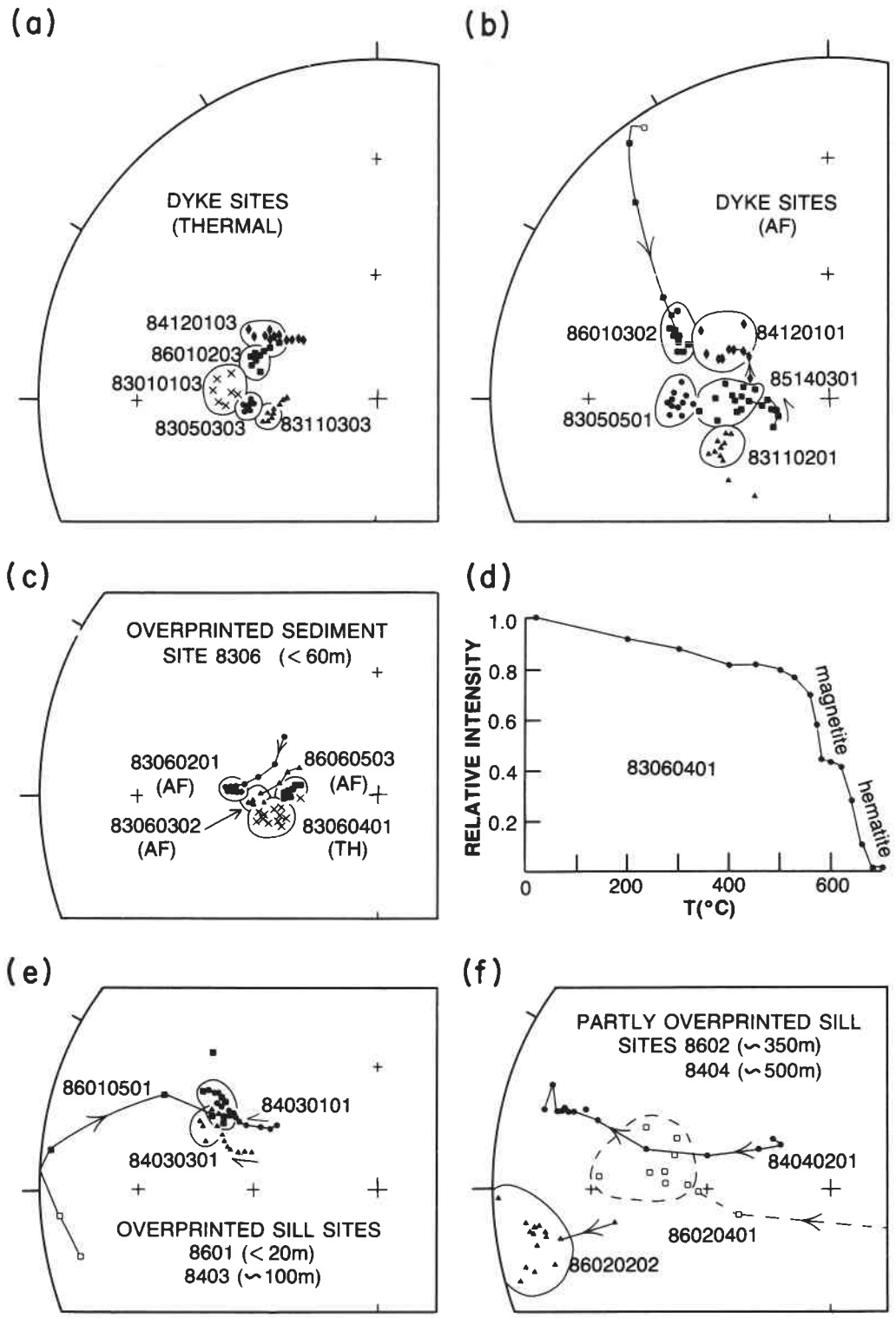


FIG. 10. Typical results with respect to the present horizontal for the northeast-trending dyke and overprinted host rocks: (*a*) thermal demagnetization of dyke specimens from several sites; (*b*) AF demagnetization of dyke specimens; (*c*), (*d*) thermal demagnetization of Coleman sediment site <60 m from dyke contact; (*e*) AF demagnetization of Nipissing sill specimens at <20 m and ~100 m from dyke contact; (*f*) AF demagnetization of Nipissing sill specimens at ~350 and ~500 m from dyke contact.

(8305 and 8311) carry only the N2 component (Figs. 10*a*, 10*b*). Coleman sediments within 60 m of the dyke contact (site 8306) also have a stable N2 magnetization (Fig. 10*c*). The remanence at sites 8312 and 8307 (100 and 200 m from the contact respectively) is less stable but directions show a broad scatter about the N2 mean. Farther from the dyke, Coleman redbeds were sampled at site 8310 (900 m from the contact) and the basal conglomerate at sites 8313 and 8314 (600 m from contact). Results for these sites were described in an earlier section. The redbeds of site 8310 carry a stable magnetization

TABLE 1. Mean directions for N3 sites

Site	N	n	AF/temp.	D (°)	I (°)	k	α_{95} (°)
				NIPISSING SILLS			
8339	5	5	80 mT	269.8	−61.5	767	2.8
8340	5	5	70 mT	254.4	−63.3	128	6.8
8401	5	5	100 mT	264.7	−55.5	32	13.7
8405	5	5	70 mT	264.3	−58.5	802	2.7
8406	5	5	90 mT	270.7	−58.2	265	4.7
8407	4	4	25 mT	271.8	−56.1	1545	2.3
8408	5	5	40 mT	250.1	−51.3	160	6.1
8409	5	5	90 mT	272.8	−63.6	1373	2.1
8410	5	5	35 mT	284.1	−54.5	341	4.1
8411	5	4	40 mT	268.5	−54.9	96	9.4
8513	5	5	80 mT	279.1	−62.6	719	2.9
				OVERPRINTED COLEMAN SEDIMENTS			
8308	5	4	580°C	264.8	−82.4	287	5.4
8309	5	5	620°C	239.0	−77.2	110	7.3
8338	5	5	50 mT	270.7	−44.0	126	6.8
8512	5	5	20 mT	273.3	−35.8	57	10.2
N3 site mean (15 sites)				268.0	−59.0	42	6.0
Mean N3 paleopole				18.8°W	29.5°N	dm = 8.9°	dp = 6.6°

NOTES: N is the number of samples collected; n is the number of samples used in the statistical calculations; AF/temp. is the peak alternating field or temperature used to calculate the mean direction; D and I are the mean declination and inclination respectively; k is the precision parameter; α_{95} is the radius of the circle of 95% confidence; dm and dp are the semi-axes of the oval of 95% confidence.

TABLE 2. Mean directions for N2 sites

Site	N	n	AF (mT)	D (°)	I (°)	k	α_{95} (°)
				NORTHEAST-TRENDING DYKE			
8301	5	5	50	271.1	54.2	121	7.0
8305	5	5	80	275.6	54.4	106	7.5
8311	5	5	15	265.6	62.2	196	5.5
8412	4	4	15	293.7	59.1	410	4.5
8514	5	4	80	285.4	64.7	40	14.7
8601[a]	3	3	30	290.0	48.5	575	5.1
				OVERPRINTED NIPISSING DIABASE			
8403	4	4	20	293.6	50.7	681	3.5
8601[a]	3	3	35	297.1	50.0	3260	2.2
				OVERPRINTED COLEMAN SEDIMENTS			
8302	5	5	15	298.6	78.7	65	9.5
8303	5	5	15	301.1	70.8	26	15.1
8304	5	5	90	296.0	62.0	23	16.1
8306	5	5	15	268.5	62.5	434	3.7
8307	5	5	30	263.6	71.7	57	10.2
8312	5	5	70	253.8	54.4	48	11.1
N2 site mean (14 sites)				282.0	61.1	48	5.8
Mean N2 virtual geomagnetic pole				145.8°W	36.9°N	dm = 8.9°	dp = 6.8°

NOTE: Symbols and abbreviations are defined in Table 1.

[a]This sampling locality contains both dyke and contact samples in a Nipissing sill. It is treated as two sites.

with a shallow northwest direction (Fig. 4), while the conglomerate cobbles at sites 8313 and 8314 are stably but randomly magnetized (Fig. 5). Clearly, these sites have not been overprinted during dyke intrusion, whereas sites 8306, 8307, and 8312 close to the dyke have been overprinted. These results constitute a positive baked contact test and demonstrate that the N2 dyke magnetization is primary.

At the second locality the northeast dyke cuts Nipissing Dia-

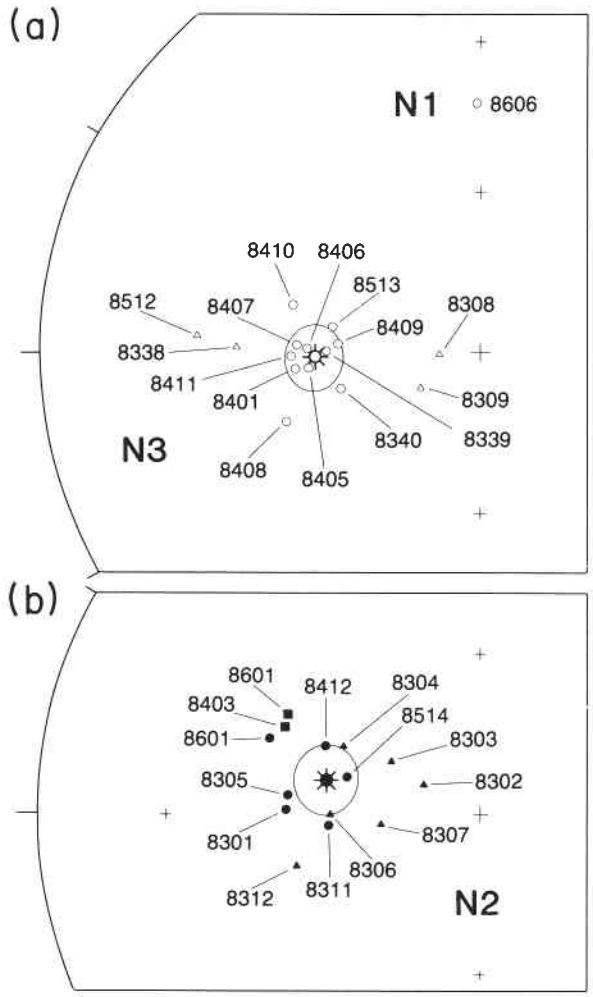


FIG. 11. Mean directions of magnetization plotted with respect to the present horizontal: (*a*) Nipissing sill sites (circles) and overprinted sediment sites (triangles); (*b*) northeast-trending dyke sites (circles), overprinted Nipissing sill sites (squares), and overprinted sediment sites (triangles). The means of the site magnetization directions are shown as stars with their circle of 95% confidence.

base (see Fig. 3). Dyke sites 8412, 8514, and 8601 have an N2 magnetization (Figs. 10*a*, 10*b*). Site 8601 also contains Nipissing sill samples collected within 20 m of the dyke contact. They carry only an N2 component (Fig. 10*e*). At site 8403, about 100 m from the contact the N2 direction is also obtained, but a slight swing towards N3 in all specimens (Fig. 10*e*) may indicate the survival of a very small N3 component at this distance from the dyke.

Sill sites 8602 (~350 m from contact) and 8404 (~500 m) carry superimposed N2 and N3 components (Fig. 10*f*). At 8404 the N2 component dominates, although systematic swings toward the N3 direction indicate the presence of a substantial N3 component with high coercivities and blocking temperatures. The persistence of the N2 magnetization as the dominant component at such a large distance from the dyke may be due to the presence of a second dyke or an offshoot of the main dyke, since an irregular contact between diabase and metamorphosed gabbro of the sill is observed at this site (see petrological description of this site below).

It is only at 800 m from the dyke (site 8405) that N3 becomes the dominant remanence (see Figs. 3, 6*c*). However, even here, as in many other N3 sill sites, there is evidence in some samples of an N2 component with lower coercivities and blocking temperatures than those of the N3 component. The persistence of the N2 component far from the dyke may represent a mild regional overprint at the time of dyke intrusion. It makes the interpretation of the baked contact less certain than in the case of the dyke intruding Coleman sediments, which was described earlier.

Analysis

In analysing data from the Englehart area, no attempt has been made to structurally correct the Nipissing sill or northeast dyke remanences. Such corrections are minor and would not greatly affect the directions described in this paper. In addition, structural correction may be inappropriate, since the variations in dip observed in nearby sediments are thought to be controlled by Archean bedrock topography and differential compaction rather than by tectonism (Lovell and Ploeger 1980).

Eleven Nipissing sill sites and four Coleman sediment sites close to the sill carry a characteristic N3 magnetization (Fig. 2). Sample directions are determined as the mean of two individual specimen directions. Site mean directions (Table 1 and Fig. 11*a*) are then calculated using the AF demagnetization step ($\geq$15 mT) with the least scatter of sample directions. The N3 sill and sediment sites have a mean direction of $D = 268.0°$, $I = -59.0°$ ($k = 42$, $\alpha_{95} = 6.0°$), and a corresponding paleopole at 18.8°W, 29.5°N (dm = 8.9°, dp = 6.6°).

The single N1 Nipissing sill site (site 8606 of Fig. 2) has a mean direction of $D = 358.9°$, $I = -43.1°$ ($k = 393$, $\alpha_{95} = 4.6°$). It is included in Fig. 11*a*.

Samples at six northeast dyke sites carry the N2 direction, while samples from two Nipissing sill sites and four Coleman sediment sites are overprinted by the dyke. Site mean directions are shown in Fig. 11*b* and are listed in Table 2. The N2 sites from dyke, sill, and sediments have a mean direction of $D = 282.0°$, $I = 61.1°$ ($k = 48$, $\alpha_{95} = 5.8°$). The corresponding virtual geomagnetic pole is located at 145.8°W, 36.9°N (dm = 8.9°, dp = 6.8°). A virtual pole rather than a paleopole is indicated, since data from a single dyke and adjacent baked sediments are unlikely to average out secular variation.

No mean direction has been calculated for the N2 component at partially overprinted N3 sill sites. This is due to the difficulty in separating N2 from viscous components aligned with the present Earth's magnetic field. Nevertheless, it is clear that some specimens at N3 sites carry a lower coercivity, lower blocking temperature N2 component.

Petrography

Nipissing sills and northeast dyke

Petrographic characteristics and paleomagnetic directions at each sill and dyke site are summarized with an estimate of percentage and type of alteration in Table 3.

In the Englehart area most N3 Nipissing sites are relatively fresh two-pyroxene gabbro consisting of clinopyroxene, orthopyroxene, plagioclase, and minor biotite, amphibole, uralite, quartz, and iron–titanium oxides (up to about 2% of medium-grained magnetite–ilmenite). These sites are indistinguishable petrographically from fresh Nipissing sills that carry an N1 or N2 remanence direction and are located in other areas of the Southern Province (Buchan and Card 1985). Exceptions include the metagabbro of site 8411 and the fresh

TABLE 3. Petrographic and magnetic characteristics of intrusive rocks

Site	Rock type	Alteration %	Alteration Type	Silicate mineralogy	Oxide mineralogy	Magnetic direction
				NIPISSING SILLS		
8339	Pyroxene gabbro	2	B	Clinopyroxene, orthopyroxene, plagioclase	1% large magnetite–ilmenite grains and minor wormy intergrowth	N3
8340	Olivine gabbro	1	B	Clinopyroxene, orthopyroxene, olivine, plagioclase	1% large magnetite–ilmenite grains plus fine-grained oxide replacing olivine	N3 + N2
8401	Pyroxene gabbro	5	Di	Orthopyroxene, clinopyroxene, plagioclase, amphibole, biotite	1% medium-grained magnetite–ilmenite	N3
8402	Granophyric metagabbro	70	C	Uralitized pyroxenes, saussuritized plagioclase, granophyre, biotite	2–3% large grains intergrown with silicates	Unstable
8403	Metagabbro (~100 m from dyke)	100	C	Uralitized amphiboles, saussurite, minor granophyre	2–3% large polyphase grains intergrown with silicates	N2
8404	Pyroxene gabbro and metagabbro (~500 m from dyke)			Two phases—coarse metagabbro in sharp contact with fine pyroxene gabbro	1–2% oxides	N2 + N3
8405	Pyroxene gabbro	1	B	Clinopyroxene, orthopyroxene, plagioclase, minor granophyre	2% medium-grained magnetite–ilmenite	N3
8406	Pyroxene gabbro	1	B	As above	1% as above	N3
8407	Pyroxene gabbro	5	B	As above with minor uralite, biotite	1–2% as above	N3
8408	Altered granophyric gabbro	30	Dii	Partly uralitized pyroxenes, saussuritized plagioclase, abundant granophyre	3% large-grained magnetite–ilmenite	N3
8409	Pyroxene gabbro	1	B	Clinopyroxene, orthopyroxene, plagioclase, minor biotite, uralite	1% medium-grained magnetite–ilmenite	N3
8410	Pyroxene gabbro	5	B	As above	As above	N3
8411	Metagabbro	50	C	Uralitized pyroxenes, saussuritized plagioclase, granophyre	1–2% medium-grained magnetite–ilmenite	N3
8513	Pyroxene gabbro	1	C	Clinopyroxene, orthopyroxene, plagioclase, quartz, granophyre biotite	1% medium-grained magnetite–ilmenite	N3
8601	Pyroxene gabbro (<20 m from dyke)	5	C	Clinopyroxene, orthopyroxene, plagioclase, amphibole, biotite, minor quartz	1% large grains	N2
8602	Metagabbro (~350 m from dyke)	100	C	Uralitic amphiboles, chlorite, epidote	1% fine-grained oxides and leucoxene aggregates	N2 + N3
8606	Olivine gabbro	21	C	Olivine, clinopyroxene, plagioclase, minor biotite, chlorite, amphibole	1% magnetite–ilmenite grains; some fine-grained aggregates around olivine	N1
				NORTHEAST-TRENDING DYKE		
8301	Altered gabbro	70	C	Uralitized pyroxenes, saussuritized plagioclase	1% wormy intergrowth	N2
8305	Altered gabbro	20	Dii	Saussuritized plagioclase, clinopyroxene	1% medium-grained magnetite–ilmenite	N2
8311	Altered gabbro	50	C	Uralitized pyroxenes, saussuritized plagioclase	1–2% very coarse, magnetite lamellae in leucoxene after ilmenite (?)	N2
8412	Pyroxene gabbro	10	Dii	Saussuritized plagioclase, pyroxenes	1–2% medium-grained magnetite–ilmenite	N2
8514	Olivine diabase	Minor		Clinopyroxene, olivine, plagioclase, minor biotite, chlorite, sericite	1% large polyphase grains and fine aggregates around olivine	N2
8601	Olivine diabase	20	Di	Partly uralitized pyroxenes (cpx), plagioclase, amphiboles, partly altered olivine grains	1–2% fine polyphase aggregates	N2

NOTES: Percentage alteration is based on silicate mineralogy. Alteration type: B, deuteric, marginal alteration of individual mineral grains; C, pervasive, alteration of pyroxenes to uralitic amphiboles and saussuritization of plagioclase; Di, selective, pyroxenes completely or partly altered, plagioclase mainly fresh; Dii, selective, plagioclase completely or partly altered, pyroxene mainly fresh.

TABLE 4. Petrographic and magnetic characteristics of Coleman sediments

Site	Distance to intrusive contact	Sample description	Opaque mineralogy	Magnetic direction	Remanence carrier (blocking temperature)
		SERIES 1: UNDER SILL, WEST END OF WENDIGO LAKE			
8308	150 m	Reddish-brown siltstone	Hematite (abundant plates, dust patches, and anhedral grains)	N3	Hematite (580–680°)
8309	100 m	Grey siltstone with chlorite spotting	Abundant hematite (irregular skeletal grains, some plates); rare magnetite (large cubes, grains)	N3	Hematite (620–680°)
8338	15 m	Dark grey siltstone with chlorite spotting, coarse sericite, welded quartz, epidote, metamorphic texture	Abundant magnetite (cubes)	N3	Magnetite (450–580°)
		SERIES 2: NEAR DYKE, MIDDLE OF WENDIGO LAKE			
8310	900 m	Brown siltstone with epidote	Hematite (small plates, rare as anhedral skeletal grains and dust)	C	Hematite (620–700°)
8307	200 m	Red siltstone	Hematite (abundant interstitial dust and coarse anhedral grains)	N2	Hematite and magnetite
8312	100 m	Red siltstone	Hematite (very abundant interstitial dust and altered clasts, some plates)	N2	?
8306	<60 m	Reddish-grey siltstone with chlorite spotting and hematite blotches	Hematite (fine-grained flakes and dust restricted to blotches); magnetite (cubes)	N2	Hematite (620–680°C); magnetite (500–580°C)
		SERIES 3: NEAR DYKE, NORTH ARM OF WENDIGO LAKE			
8304	~150 m	Brown siltstone	Hematite (dusting, euhedral flakes, anhedral grains)	N2	Hematite (600–660°C?)
8303	~70 m	Dark brownish-grey siltstone with chlorite spotting	Diagenetic hematite (clusters of plates, grains)	N2	Hematite?
8302	<10 m	Greyish-green sandstone with welded quartz, chlorite spotting, and epidote	Pyrite; possible fine-grained magnetite	N2	Magnetite (<600°C)
		SERIES 4: NEAR SILL, EAST OF WENDIGO LAKE			
8510	5 km	Grey coarse-grained sandstone with Ti_2O_2 granules	Pyrite (poikilotopic)	Unstable	—
8511	2 km	Greyish-brown siltstone–mudstone	Pyrite; possible hematite as fine anhedral grains	Unstable	—
8512	<50 m	Grey fine-grained sandstone with welded texture, TiO_2 grains	Rare unidentified opaque grains, fine dusting of hematite in feldspars	N3	Magnetite (500–580°C)

olivine gabbro of site 8340. Olivine gabbro is not typical of Nipissing sills from other areas, except locally in sills near Cobalt (Hriskevitch 1968).

The single N1 site from the Englehart area (site 8606) is also composed of olivine gabbro that is partly altered. The granophyric metagabbro of site 8402 is unstably magnetized. Sill sites lying close to the northeast dyke are remagnetized in the direction of the dyke and usually consist of pervasively altered metagabbro.

The dyke itself, which carries the N2 direction at all sites, has a hypidiomorphic equigranular texture and consists of variably uralitized pyroxenes, saussuritized plagioclase, and minor biotite, chlorite, and iron–titanium oxides. Fresh clinopyroxene and olivine are present in several samples. Chemical analyses of samples from each dyke site show that the dyke is chemically similar to Nipissing Diabase and Preissac dykes. Therefore, the northeast dyke near Englehart is tentatively correlated with the 2150 Ma Preissac dyke swarm. However, the possibility that it belongs to the much younger (1140 Ma) Abitibi swarm cannot be entirely ruled out.

Huronian sediments

Sediment sampling sites have roughly been located along lines at right angles to intrusive contacts at four localities. Series 1 consists of sites 8308, 8309, and 8338 close to the Nipissing sill at the western end of Wendigo Lake (Fig. 3); series 2 includes sites 8310, 8307, 8312, and 8306 on either side of the 150 m wide dyke at the middle of Wendigo Lake (Fig. 3); series 3 incorporates sites 8304, 8303, and 8302 near the dyke along the northern arm of Wendigo Lake (Fig. 2); series 4 includes sites 8510, 8511, and 8512 at decreasing distances from the Nipissing sill about 7 km east of Wendigo Lake (Fig. 2). Results of the petrographic study of each series are tabulated in Table 4.

Although data from series 4 are difficult to interpret, series 1–3 show consistent trends in petrographic and magnetic characteristics as the intrusive contact is approached. Hematite, which is common as fine dust or plates far from the contact, is replaced by magnetite cubes close to the intrusion. This change coincides with a change in sample colour from red to grey. Textures become welded and chlorite spotting appears.

Chlorite spotting is characteristic of the thermal aureole of the Nipissing Diabase (Chandler, in press).

The replacement of hematite by magnetite close to the intrusions is confirmed by thermal demagnetization studies. Nevertheless, the zone in which the magnetic remanence has been reset in the intrusion direction extends, in each case, well beyond the zone where magnetite has replaced hematite. This demonstrates that the magnetic overprinting near these intrusions is not simply a chemical remanent magnetization, but must be of thermal origin at sites where hematite carries the overprint remanence. The thermal nature of the overprint is important in assessing baked contact test data.

Discussion

The N3 magnetization is the dominant remanence of Nipissing sills in the Englehart area. This component has not been reported in other areas of Nipissing sills, except, perhaps, in the New Liskeard area, where Roy and Lapointe (1976) found a somewhat similar magnetization direction ($D = 271°$, $I = -26°$, $k = 14$, $\alpha_{95} = 14°$) in sites located near a fault. This component was interpreted to be a local overprint due to faulting. It is clearly important to establish whether the N3 magnetization is primary or secondary.

The widespread occurrence of the N3 magnetization in the Englehart area demonstrates that it is not simply an overprint related to faulting. It can also be shown that the Englehart N3 component is not a rotated version of the N1 component observed in Nipissing sills from other areas of the Southern Province, since this would also result in a rotation of the N1 component of the Englehart-area site 8606 and the N2 component of the northeast dyke into anomalous directions.

The N2 overprinting of Nipissing sill sites adjacent to the crosscutting diabase dyke shows that the N3 component predates intrusion of the dyke. Since the dyke is undoubtedly of either the Abitibi (~1140 Ma) or, more likely, the Preissac (~2150 Ma) swarm, the N3 magnetization can be no younger than 1150 Ma and is probably older than about 2150 Ma. This precludes, for example, an overprint at the time of the Grenville orogeny (~1000–900 Ma).

The positive baked contact test with Coleman sediments demonstrates that the Nipissing Diabase overprinted adjacent siltstones at the time of intrusion, while farther away the basal conglomerate and the siltstone site 8310 were not overprinted. As noted in the petrography section the overprinting is, at least in part, a thermoremanent magnetization. This positive baked contact test indicates that the N3 magnetization is indeed primary, acquired at the time of Nipissing intrusion. Positive baked contact tests, although of variable quality (McGlynn and Irving 1981), have also been reported for N1 and N2 Nipissing sills from other areas (Symons 1967; Morris 1979; Roy and Lapointe 1976). Thus, three ages of intrusion may be represented by the sills collectively known as the Nipissing Diabase.

Petrographic studies of Nipissing Diabase from several localities in the Southern Province (this study; Buchan and Card 1985) have failed to show any systematic differences in degree of alteration of samples that carry different paleomagnetic directions. All three Nipissing components are carried, at least at some sites, by mineralogically unaltered gabbro. This observation supports the hypothesis of three distinct ages of sills, since petrographically fresh gabbro with magnetic blocking temperatures near 580°C is unlikely to have acquired a complete magnetic overprint. Nevertheless, some sites that

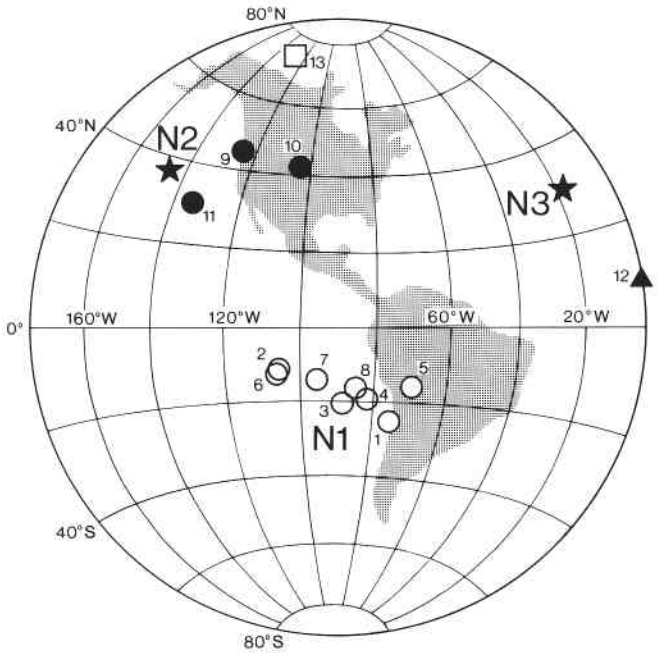


FIG. 12. Paleopoles for Nipissing intrusions and Preissac dykes. N1 Preissac pole: 1, Larochelle (1966). N1 Nipissing poles: 2, Symons (1967); 3, Symons (1970); 4, Symons (1971) recalculated by Morris (1981); 5, Patel and Palmer (1974); 6, Symons and Londry (1975); 7, Roy and Lapointe (1976); 8, Stupavsky and Symons (1982). N2 Nipissing poles: 9, Symons (1971) recalculated by Morris (1981); 10, Roy and Lapointe (1976). N2 Preissac pole: 11, Irving and Naldrett (1977) recalculated as described below. N3 Nipissing pole: 12, Roy and Lapointe (1976). Recent overprint of Nipissing intrusions: 13, Symons and Londry (1975). Stars represent the paleopole for the N3 Nipissing sills and contact rocks and the virtual geomagnetic pole for the N2 dyke and contact rocks from the present study. Note that pole 11 is calculated for sites 1–5 of Irving and Naldrett (1977). These sites belong to a 2150 Ma Preissac dyke, while other sites (6 and 7) are now known to belong to a 1140 Ma Abitibi dyke.

are petrographically fresh have suffered at least partial remagnetization (e.g., an N2 component is superimposed on an N3 component at some Englehart-area sampling sites).

As described earlier, several isotopic age determinations have been reported for Nipissing sills. These include Rb–Sr whole-rock and biotite ages of 2109–2134 Ma and a U–Pb baddeleyite age of 2219 Ma. No age has yet been reported for Nipissing sills in the study area near Englehart.

Indeed, no convincing age determination has been obtained at an N1, N2, or N3 paleomagnetic site. Andrews *et al.* (1986) collected paleomagnetic samples some distance (170 m) from the Gowganda locality where they measured the 2219 Ma baddeleyite age. They suggest that the Nipissing sill at this location belongs to the N2 set, but the preliminary paleomagnetic data on which this interpretation is based are very complex (four remanence components) and this interpretation cannot be considered conclusive.

P. Lightfoot (personal communication, 1988) has collected samples for U–Pb baddeleyite dating from some Nipissing paleomagnetic sites. Further work is planned to determine absolute ages of all three Nipissing sill magnetizations, as well as magnetizations of N1 and N2 Preissac dykes.

In Fig. 12 published paleopoles for the Nipissing sills and Preissac dykes (thought to be of roughly equivalent age) are

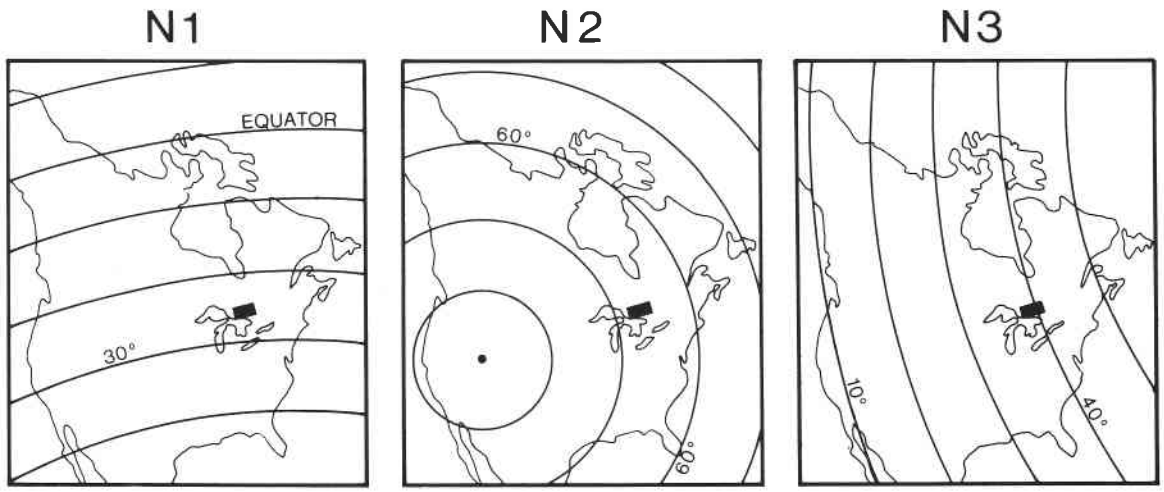


FIG. 13. Paleolatitudes of North America at the time of N1, N2, and N3 remanence acquisition. The Southern Province is shown as a black rectangle.

compared with results obtained in the present study. Two major groupings (N1 and N2) are evident. The N3 paleopole that characterizes the Nipissing sills of the Englehart area is far removed from the N1 and N2 groupings. The virtual pole for the northeast-trending dyke of the study area falls in the N2 grouping, although, as noted above, it could belong to the much younger (1140 Ma) Abitibi dyke swarm, which has a magnetic signature similar to that of N2.

The three magnetizations observed in Nipissing sills were acquired when North America was at widely divergent paleolatitudes (Fig. 13). These paleolatitudes can be used to estimate the minimum length of time over which the three Nipissing components were acquired. Let us assume that the magnetizations were acquired in a sequence that minimizes north–south movement. For example, the sequence N2–N3–N1 or N1–N3–N2 with all three paleolatitudes in the same hemisphere would require a paleolatitudinal change of ~40°. If we then assume a north–south drift rate of 10 cm/a, a minimum of ~50 Ma would be necessary for acquisition of all three Nipissing components. Thus, the age difference implied by Nipissing paleolatitudes should be resolvable with current high-precision dating techniques.

Conclusions

The following conclusions can be drawn from the study of Nipissing sills and associated rocks of the Englehart area.

(1) The characteristic magnetization direction (N3) of Nipissing Diabase in the Englehart area is unlike either of the Nipissing magnetization directions (N1 and N2) usually reported from other areas. A single site in the sampling area carries the N1 component.

(2) A positive baked contact test between the N3 Nipissing sills and the Coleman sediments demonstrates the primary nature of the N3 magnetization in the sills. Coleman argillite within 150 m of the sills carries an N3 overprint, while mafic cobbles of a Coleman basal conglomerate and a Coleman siltstone site farther away are not remagnetized. Petrographic study of Coleman sediments in the contact zone suggests that overprinting is, at least in part, of thermal origin.

(3) A 150 m wide, northeast-trending, diabase dyke cuts Coleman sediments and Nipissing sills in the northern portion of the sampling area. It carries a remanence direction similar to the N2 Nipissing direction observed in other areas. The dyke probably belongs to the N2 Preissac swarm (~2150 Ma), but correlation with the much younger Abitibi dykes (~1140 Ma), which also carry a direction similar to N2, cannot be ruled out.

(4) The N2 magnetization of the dyke is primary, as demonstrated by the positive baked contact test with adjacent Coleman sediments.

(5) The N2 magnetization is also observed as a lower coercivity, lower blocking temperature component superimposed on a harder N3 component at some Nipissing sill sites, even when they are located far from the dyke. This N2 component may represent a mild regional overprint that was acquired at the time of dyke intrusion.

(6) The Nipissing Diabase of most N3 sites in the Englehart area is petrographically fresh. Nevertheless, the presence of two magnetization components (N2 and N3) indicates that some sites have suffered partial overprinting.

(7) Positive baked contact tests for N1 Nipissing sills (Symons 1967; Morris 1979), N2 Nipissing sills (Roy and Lapointe 1976), and N3 Nipissing sills (this study) suggest that three distinct ages of Nipissing intrusion may be present. That the N1 and N2 sills may represent different ages of intrusion was previously suggested by Morris (1979) and Stupavsky and Symons (1982). This interpretation is supported by the relative freshness of most N1, N2, and N3 sites (Buchan and Card 1985; this study).

(8) The differences in age of N1, N2, and N3 Nipissing sills should be resolved by precise U–Pb isotopic age determinations at N1, N2, and N3 paleomagnetic sites using minerals such as baddeleyite or zircon.

(9) Most Coleman redbed sites are magnetically overprinted by either the Nipissing sills or the northeast-trending dyke, rendering it impossible to establish a primary paleopole for the Coleman Member at this locality. Nevertheless, a positive conglomerate test indicates the potential for obtaining primary paleopoles from Coleman redbed units if they are far removed from the influence of dykes or sills.

Acknowledgments

The authors wish to thank L. Madore, J. F. Marier, M. Pyne, and T. West for assistance in the field and labora-

tory. K. Christie, R. Ernst, P. Lightfoot, and W. Fahrig provided helpful discussions, K. Christie assisted with statistical analyses of the data, and T. West drafted many of the figures. The manuscript benefited from reviews by A. Davidson, D. Dunlop, and an anonymous referee.

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